ECE 340: Semiconductor Devices

Lecture 8: Carrier Concentrations

Prof. Hyunseok Kim

Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

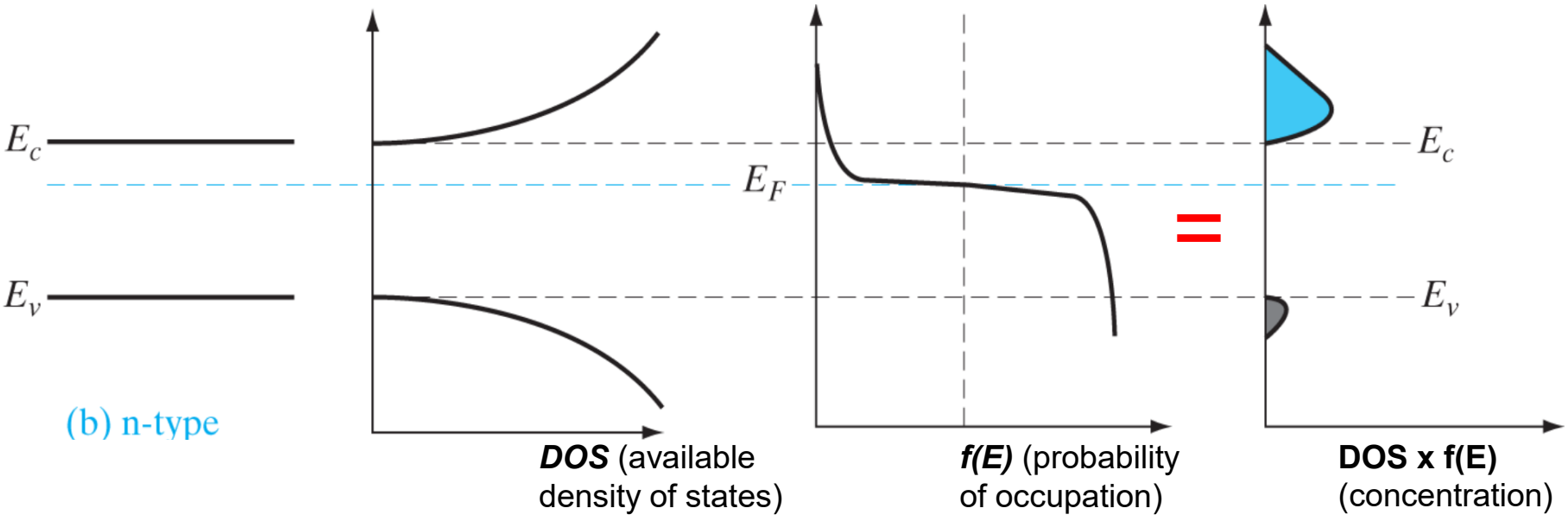
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(Recap) calculating carrier density

- Integration of [The number (density) of states at a given energy] x [the probability a state is filled at that energy]

$$n_0 = \int_{E_c}^{\infty} f(E) N_c(E) dE$$



$$p_0 = \int_{-\infty}^{E_v} [1 - f(E)] N_v(E) dE$$

(Recap) Fermi-Dirac Distribution

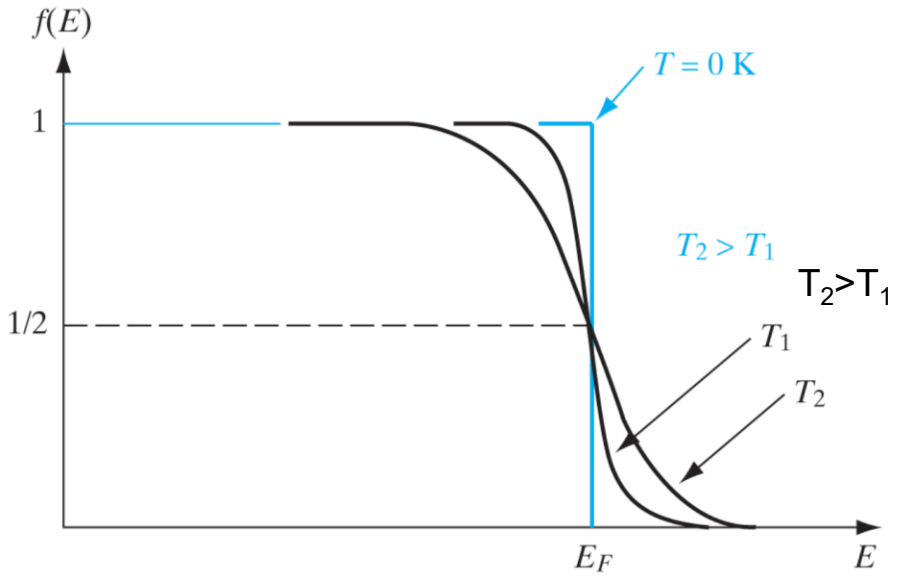
- The probability that an available energy state at E will be occupied by an electron at temperature T (temperature in Kelvin) is given by the **Fermi-Dirac Distribution Function**
- Fermi-Dirac distribution function:

$$f(E) = \frac{1}{1 + e^{(E - E_f)/kT}}$$

E_f : Fermi Level

k : Boltzmann Constant

$k = 8.62 x 10^{-5}\, eV / K$

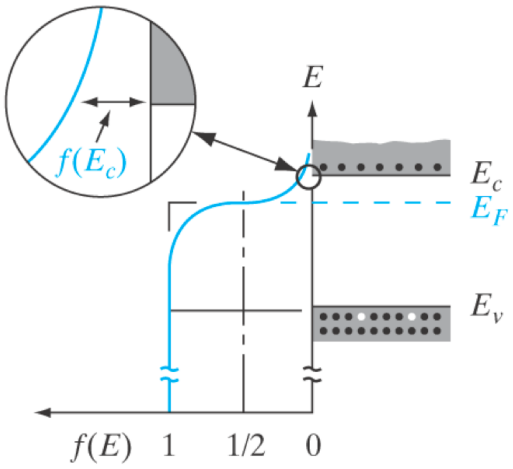


$$f(E_C) = \frac{1}{1 + e^{(E_C - E_f)/kT}} \sim e^{-(E_C - E_f)/kT}$$

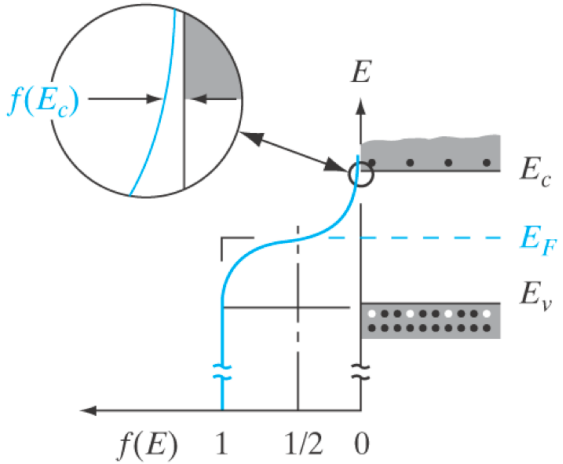
(non-degenerate case)

(Recap) Fermi Level Versus Doping

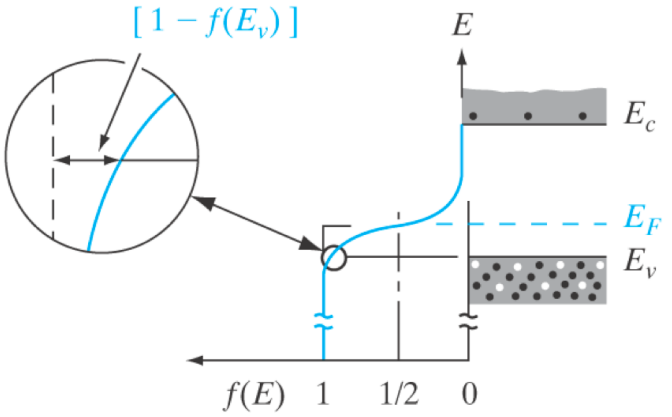
n-Type



Intrinsic



p-Type



Remember: At $T = T_1$:

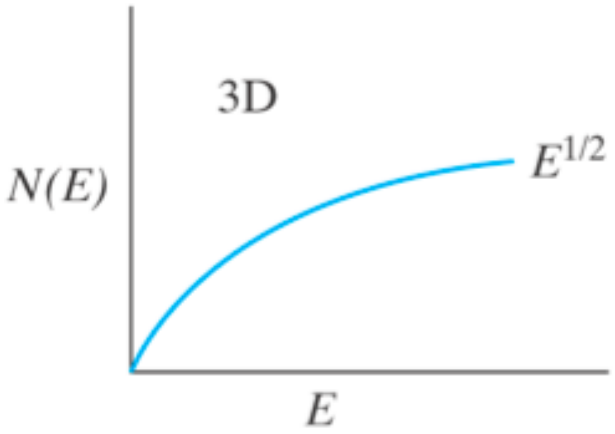
$$f(E) = \frac{1}{1 + e^{(E - E_f)/kT_1}}$$

$$1 - f(E) = \frac{e^{(E - E_f)/kT_1}}{1 + e^{(E - E_f)/kT_1}} = \frac{1}{1 + e^{(E_f - E)/kT_1}}$$

(Recap) Density of States

DoS calculated

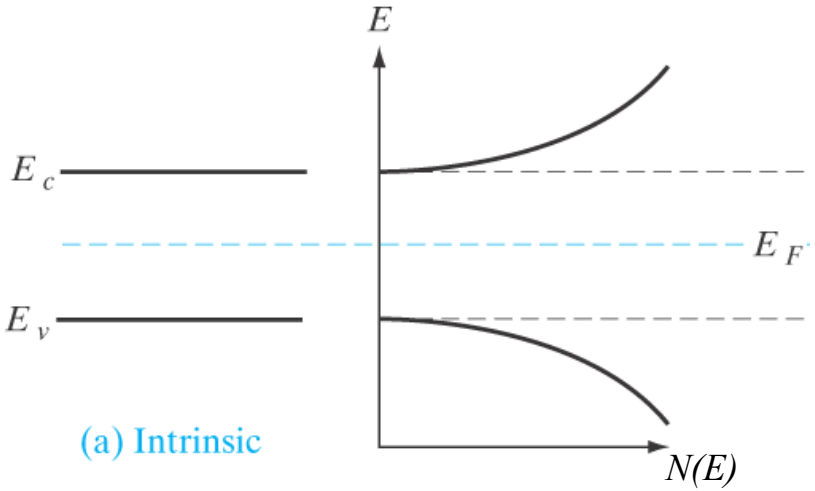
$$N(E) = \frac{dN'(E)}{dE}\frac{1}{L^3} = \frac{1}{2\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{3/2}\sqrt{E}$$



DoS of electron in conduction band

$$N_c(E) \propto \sqrt{E - Ec}$$

(E-E$_c$) instead of *E* because the energy relative to the band edge (*E$_c$*) is its kinetic energy within the band



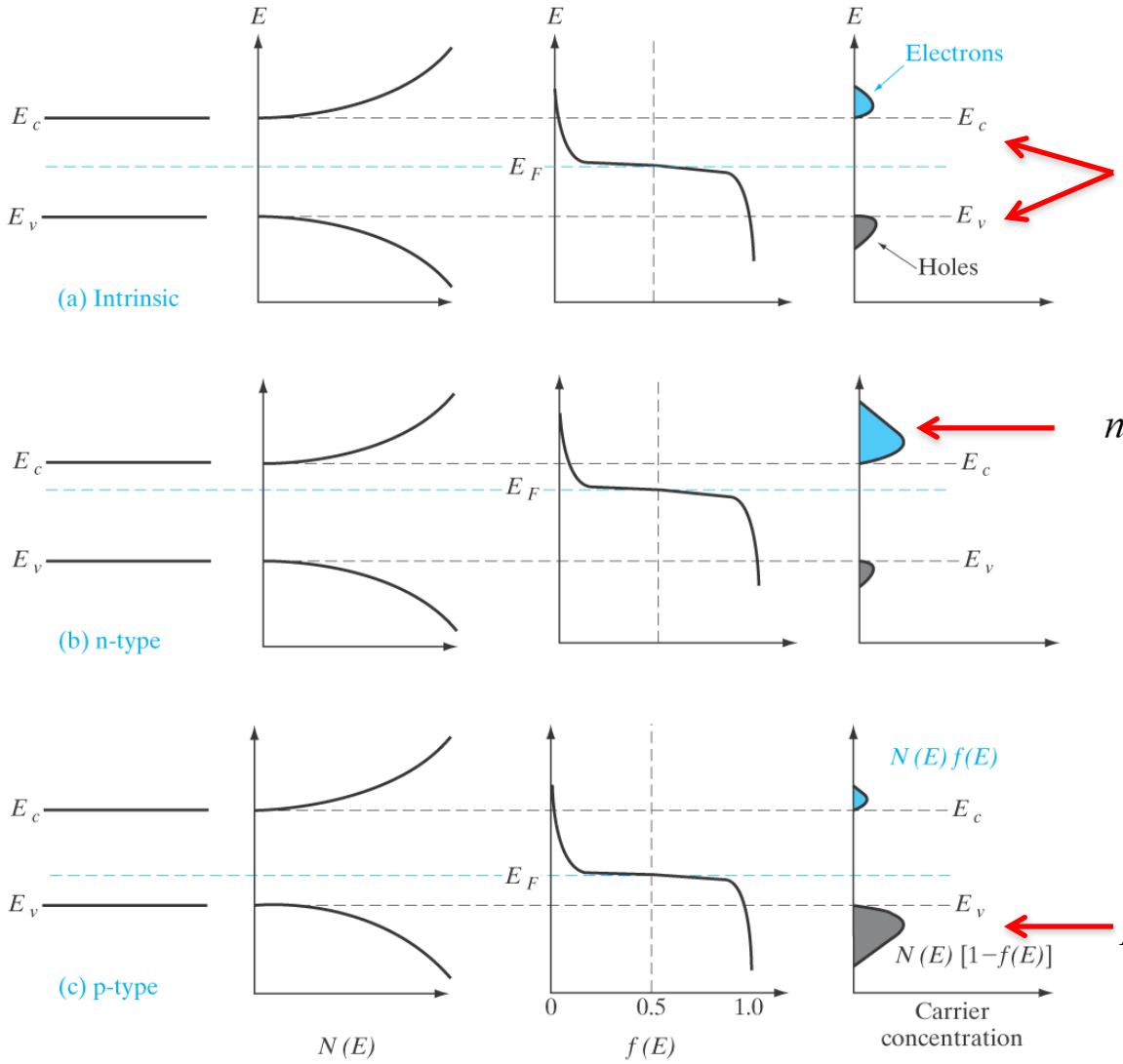
DoS of hole in valence band

$$N_v(E) \propto \sqrt{E_v - E}$$

Equilibrium Electron and Hole Concentrations

Carrier Concentration at Thermal Equilibrium

The Fermi Function Multiplied by the Density of States



n_i, p_i: intrinsic concentration

Conduction band DOS

$$n_i = \int_{E_c}^{\infty} f(E) N_c(E) dE$$

$$= p_i = \int_{-\infty}^{E_V} [1 - f(E)] N_V(E) dE$$

Probability the state is empty

Valence band DOS

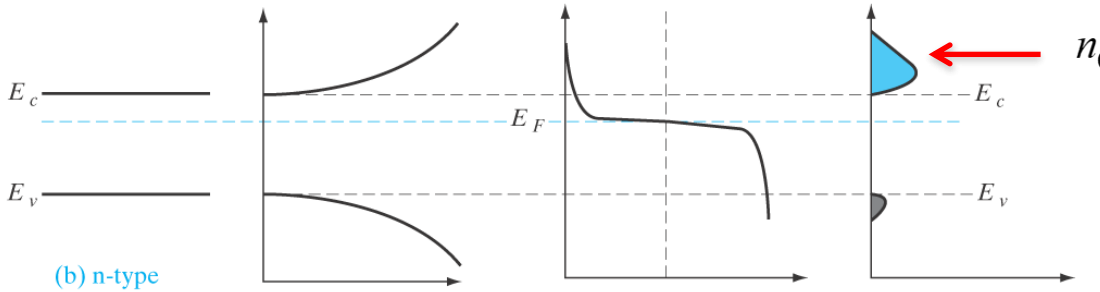
$$n_0 = \int_{E_c}^{\infty} f(E) N_c(E) dE$$

n_0, p_0: equilibrium concentration (could be intrinsic or extrinsic)

Thus, $n_0 = p_0 = n_i$ for intrinsic Si

$$p_0 = \int_{-\infty}^{E_v} [1 - f(E)] N_v(E) dE$$

Carrier Concentration n_o at Equilibrium



$$n_0 = \int_{E_c}^{\infty} f(E) N_c(E) dE$$

$$= \int_{E_c}^{\infty} \frac{1}{1+e^{(E-E_f)/kT}} \bullet \frac{\sqrt{2}}{\pi^2} \left(\frac{m^*}{\hbar^2} \right)^{3/2} E^{1/2} dE$$

$$f(E_c) = e^{-(E_c-E_f)/kT} \qquad if \qquad e^{(E_c-E_f)/kT} >> 1$$

$$n_o = \int_{E_c}^{\infty} e^{-(E-E_f)/kT} \bullet \frac{\sqrt{2}}{\pi^2} \left(\frac{m^*}{\hbar^2} \right)^{3/2} E^{1/2} dE = 2 \left(\frac{2\pi m_n^* kT}{h^2} \right)^{3/2} e^{(E_F-E_C)/kT}$$

$$= N_C(T) e^{-(E_C-E_F)/kT}$$

$$n_o = N_c \bullet e^{-(E_c-E_f)/kT}$$

where $$N_c(T) = 2 \left(\frac{2\pi \bullet m_n^* kT}{h^2} \right)^{3/2}$$

Similarly, $$p_o = N_v \bullet e^{-(E_f-E_v)/kT}$$

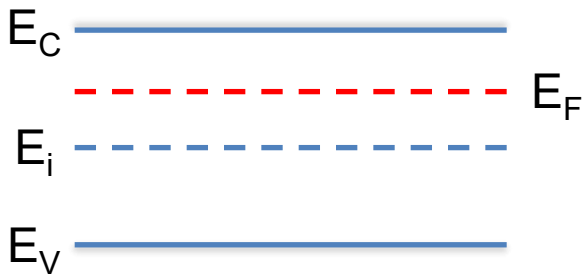
where $$N_v(T) = 2 \left(\frac{2\pi \bullet m_p^* kT}{h^2} \right)^{3/2}$$

Complete solution : Appendix IV

Non-Degenerate Case: Concentration Product and Intrinsic Concentration

Non-degenerate: E_F far enough from E_C or E_V (>3kT) for the approximation to be valid.

$$f(E_C) = \frac{1}{1+e^{(E_C-E_f)/kT}} \sim e^{-(E_C-E_f)/kT}$$



$$n_i p_i = N_c e^{-(E_c-E_i)/kT} \bullet N_v e^{-(E_i-E_v)/kT} = N_c N_v e^{-(E_c-E_v)/kT} = N_c N_v e^{-E_g/kT}$$

$$n_o p_o = N_c e^{-(E_c-E_f)/kT} \bullet N_v e^{-(E_f-E_v)/kT} = N_c N_v e^{-(E_c-E_v)/kT} = N_c N_v e^{-E_g/kT}$$

$$\text{therefore} \quad n_o p_o = n_i p_i = n_i^2$$

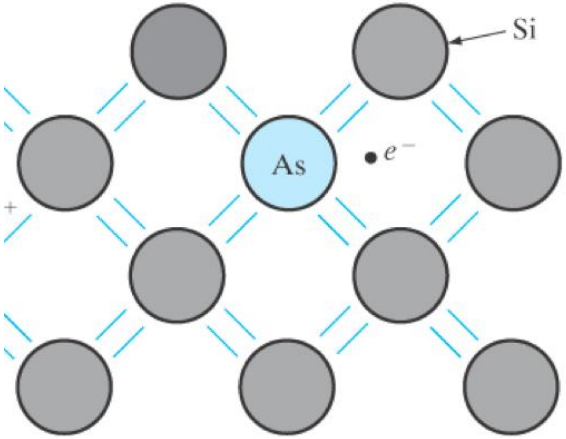
$$\text{and} \quad n_i^2(T) = N_c N_v \bullet e^{-(E_g)/kT}$$

$$n_i(T) = \sqrt{N_c N_v} \bullet e^{-(E_g)/2kT}$$

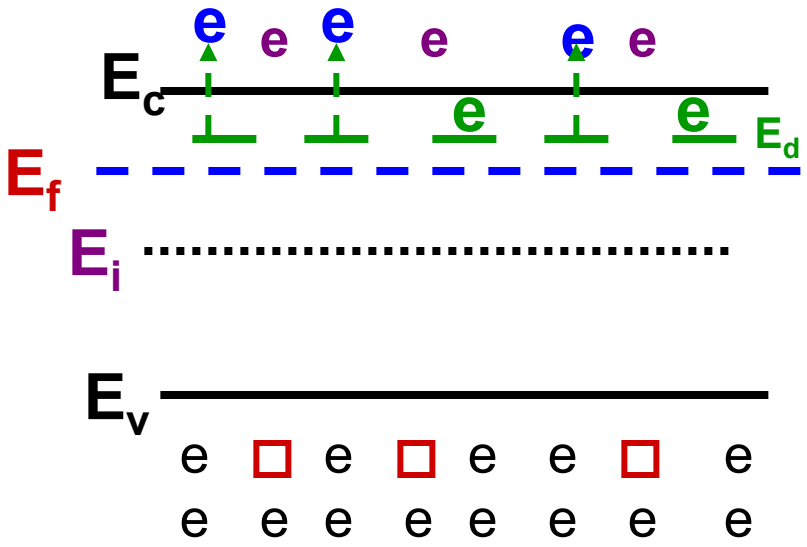
- The intrinsic carrier concentration depends upon
  - Bandgap, temperature, electron effective mass (through N_c), and hole effective mass (through N_v)
- E_i is at midgap only if $N_c = N_v$ (equal electron and hole effective masses)

Electron Concentration n_o in a n-type Semiconductor at Equilibrium

Donor concentration N_D



$n_o(K)=N_D$, majority carrier



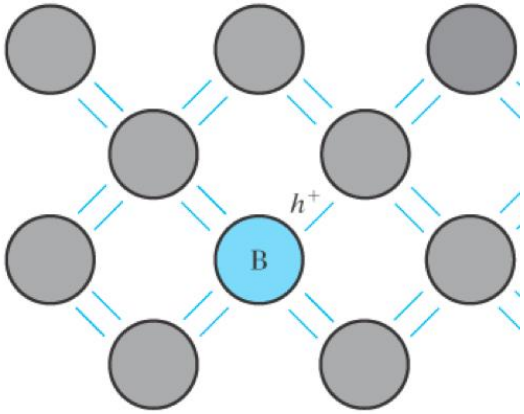
$p_o(K)=n_i^2/n_o$, minority carrier

We know $$n_o = N_c \bullet e^{-(E_c - E_f)/kT}$$ → (E_c-E_f) can be derived.

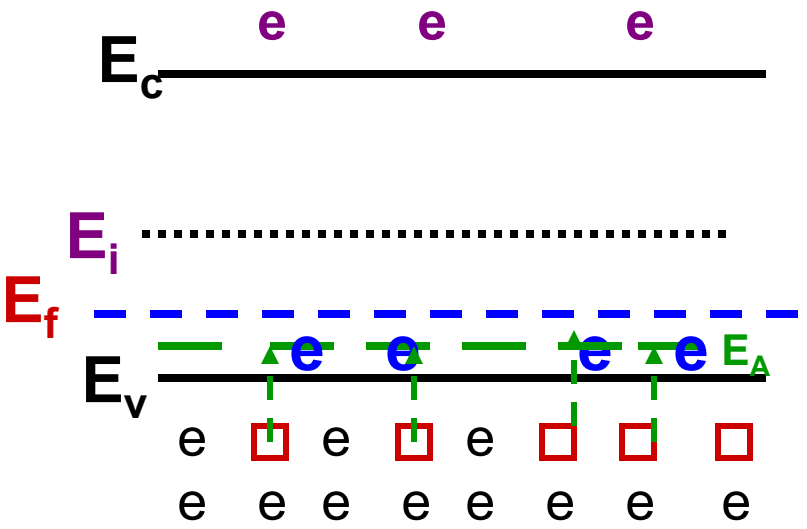
where $$N_c(T) = 2\left(\frac{2\pi \bullet m_n^* kT}{h^2}\right)^{3/2}$$

Hole Concentration p_o in a p-type Semiconductor at Equilibrium

Acceptor concentration N_A



$n_o(K) = n_i^2/p_o$, minority carrier



$p_o(K) = N_A$, majority carrier

We know $p_o = N_v \bullet e^{-(E_f - E_v)/kT}$

→ $(E_f\text{-}E_v)$ can be derived.

where $N_v(T) = 2\left(\frac{2\pi \bullet m_p^* kT}{h^2}\right)^{3/2}$

Approximating the Fermi Integral: Further Comments

$$f(E_c) = \frac{1}{1+e^{(E_c-E_f)/kT}}$$

$$f(E_c) = e^{-(E_c-E_f)/kT} \quad if \quad e^{(E_c-E_f)/kT} >> 1$$

- The approximations to the Fermi integral are valid for non-degenerate semiconductors
  - In a non-degenerate semiconductor, the Fermi Level is well within the forbidden gap (several kT away in energy)
  - From a practical perspective, the implication is that the doping density is not too high
  - For Si at room temperature:
    - $N_D < 1.6 \times 10^{18}$ cm^{-3}
    - $N_A < 9.1 \times 10^{17}$ cm^{-3}

Temperature	kT
-40° C (233K)	0.0201 eV
27° C (300K)	0.0259 eV
127° C (400K)	0.0345 eV

- The common rule of thumb is that $(E_c - E_f) > \sim 3kT$ for the approximation to be valid
  - In this case, $e^{(Ec-Ef)/kT} > 20$

The Case of Degenerate Doping

$$n = \int_{E_c}^{\infty} D_C(E) f(E) dE$$

$$D_C(E) = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{\frac{3}{2}} \sqrt{E - E_C}$$

so

$$n = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{\frac{3}{2}} \int_{E_c}^{\infty} \frac{\sqrt{E - E_C}}{1 + e^{[(E-E_F)/kT]}} dE$$

Defining $\eta = (E_F - E_C)/kT$ and $\xi = (E - E_C)/kT$

$$n = \frac{1}{2\pi^2}\left(\frac{2m_e^* kT}{\hbar^2}\right)^{\frac{3}{2}} \int_{0}^{\infty} \frac{\sqrt{\xi}}{1 + e^{(\xi-\eta)}} d\xi$$

Using the Fermi-Dirac Integral, this becomes

$$n = 2\left(\frac{2\pi m_e^* kT}{h^2}\right)^{\frac{3}{2}} F_{1/2}(\eta) = N_C F_{1/2}(\eta)$$

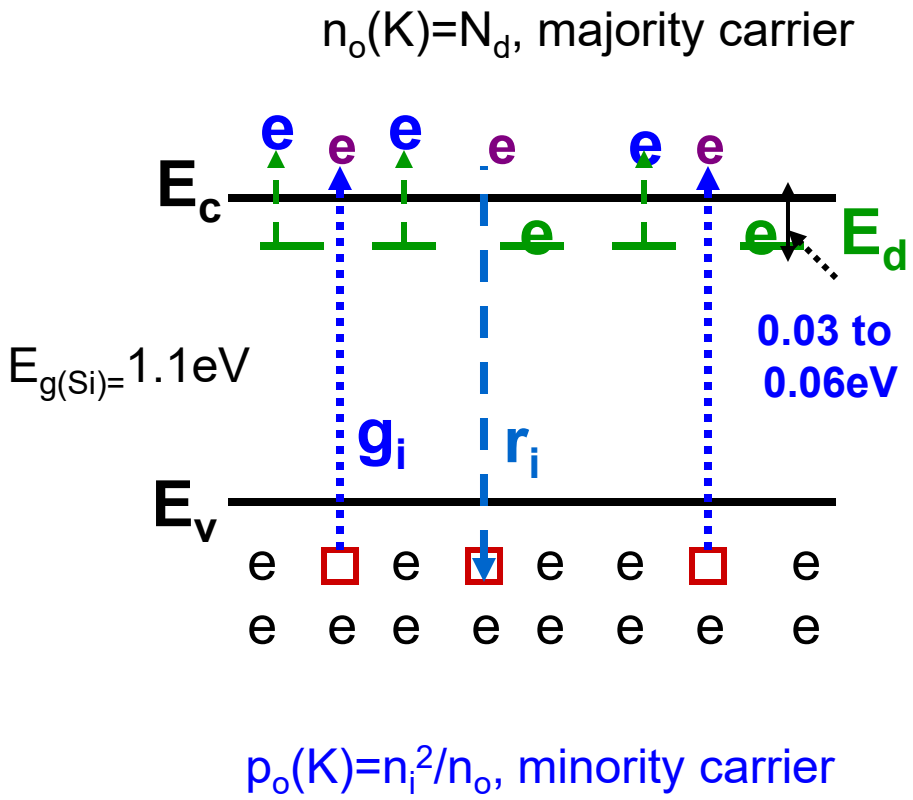
$F_{1/2}(\eta)$ is a tabulated function

- The full expressions for the density of states and Fermi distribution need to be used in the case of degenerate doping
- The integral is simplified using the Fermi-Dirac integral, which is available in tabulated form (look-up table)

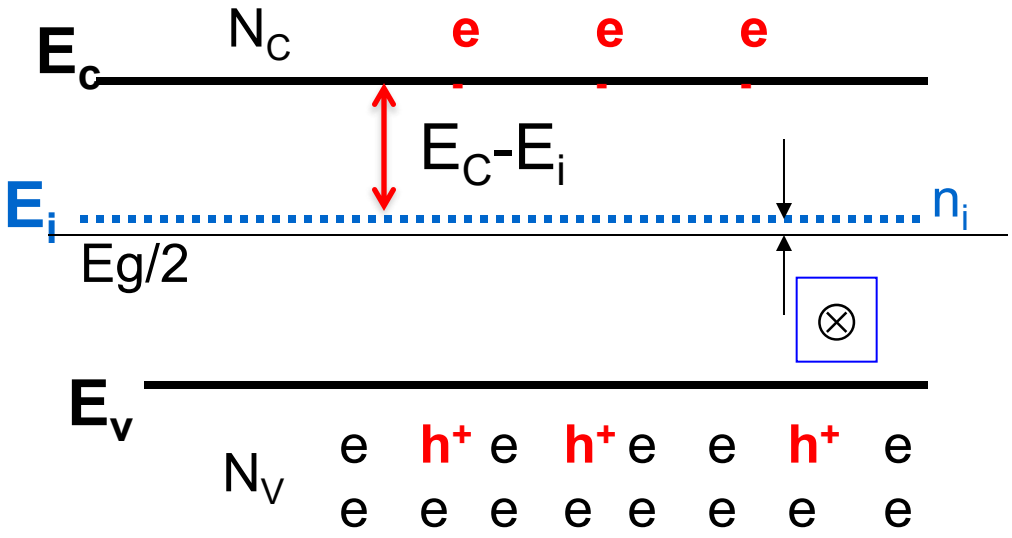
Minority Carrier Concentration in Equilibrium

- Why is the minority carrier concentration smaller than n_i?
- In an intrinsic material, recall that $n=p=n_i$
- These values are determined by the generation and recombination rates, which must be equal in equilibrium
- Generation is set by the temperature
- When the semiconductor is doped, there is an abundance of majority carriers, the minority carrier thus has a greater opportunity to be involved in a recombination event
- For recombination and generation to remain balanced, the minority carrier concentration must become smaller

$$r_i = \alpha_r n_0 p_0 = g_i$$



Intrinsic Fermi Level: Not Quite Midgap



$$n_i(T) = \sqrt{n_i p_i} = \sqrt{N_c N_v} \bullet e^{-(E_g)/2kT}$$

and $$n_i(T) = N_c \bullet e^{-(E_c - E_i)/kT}$$

so

$$N_c \bullet e^{-(E_c - E_i)/kT} = \sqrt{N_c N_v} \bullet e^{-(E_g)/2kT}$$

$$e^{(E_g)/2kT} \bullet e^{-(E_c - E_i)/kT} = \left\{ \frac{N_v}{N_c} \right\}^{1/2}$$

$$\frac{E_g}{2} - (E_c - E_i) = \Delta = kT \ln \left\{ \frac{N_v}{N_c} \right\}^{1/2}$$

$$\Delta = kT \ln \left\{ \left(\frac{m_p^*}{m_n^*} \right)^{3/2} \right\}^{1/2}$$

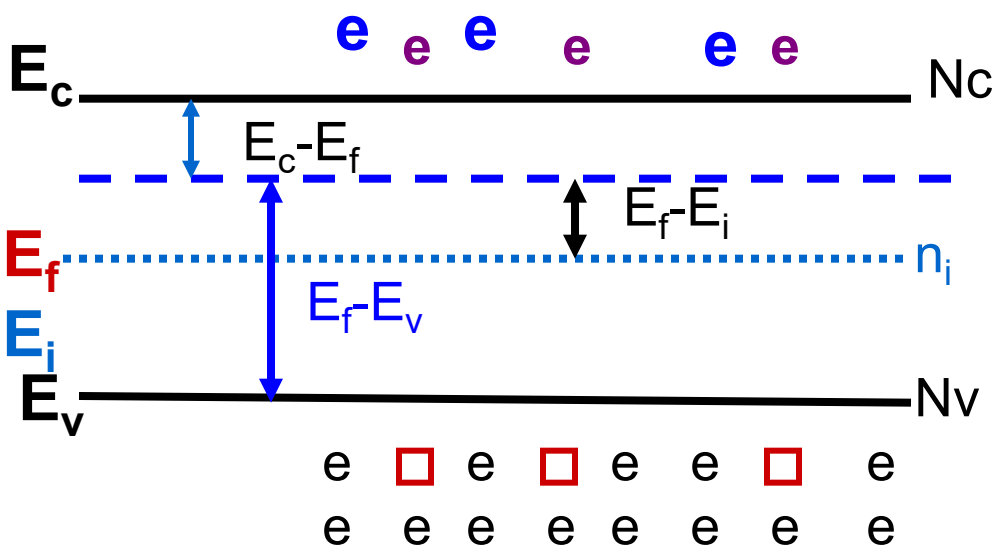
If Δ is positive, E_i is closer to E_c than midgap.

If Δ is negative, E_i is closer to E_v than midgap.

Δ is positive when $m_p^* > m_n^*$

Referencing n_o and p_o to the Intrinsic Level & Intrinsic Carrier Concentration

$n_o(K)=N_d$, majority carrier



$p_o(K)=n_i^2/n_o$, minority carrier

$$n_o = N_c \bullet e^{-(E_c-E_f)/kT}$$

$$= N_c \bullet e^{-(E_c-E_i+E_i-E_f)/kT}$$

$$= N_c \bullet e^{-\left[(E_c-E_i)-(E_f-E_i)\right]/kT}$$

$$= N_c \bullet e^{-(E_c-E_i)/kT}\, e^{(E_f-E_i)/kT}$$

but $n_i = N_c \bullet e^{-(E_c-E_i)/kT}$ so

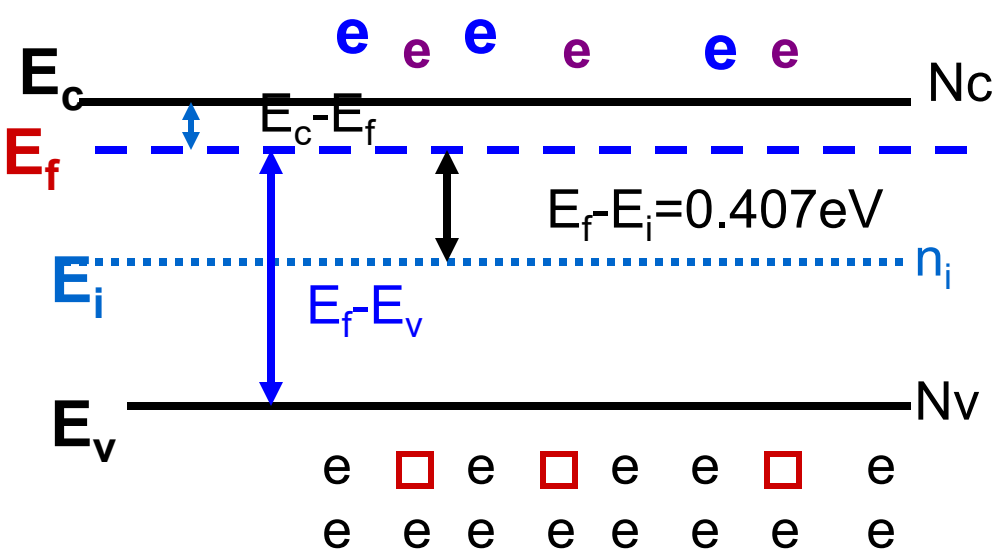
$$n_o = n_i(T) \bullet e^{(E_f-E_i)/kT}$$

Similar derivation for p_0

$$N_C = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$$

Example: Donor-Doped Silicon, $N_d=10^{17}$ cm^{-3} What is p_o at 300K? What is E_F-E_i?

$n_o(K)=N_d$ majority carrier



$p_o(K)=n_i^2/n_o$, minority carrier

$$E_c - E_i = kT \bullet \ln\left(\frac{N_c}{n_i}\right) = 0.561eV$$

$$\text{so } E_c - E_f = 0.561 - 0.407 = 0.154eV > 3kT$$

$$\text{Since} \quad N_d(As) >> n_i,$$

$$n_o = N_d = 10^{17}\,cm^{-3}$$

$$p_o = \frac{n_i^2}{n_o} = \frac{\left(1.5\times10^{10}\,cm^{-3}\right)^2}{10^{17}\,cm^{-3}}$$

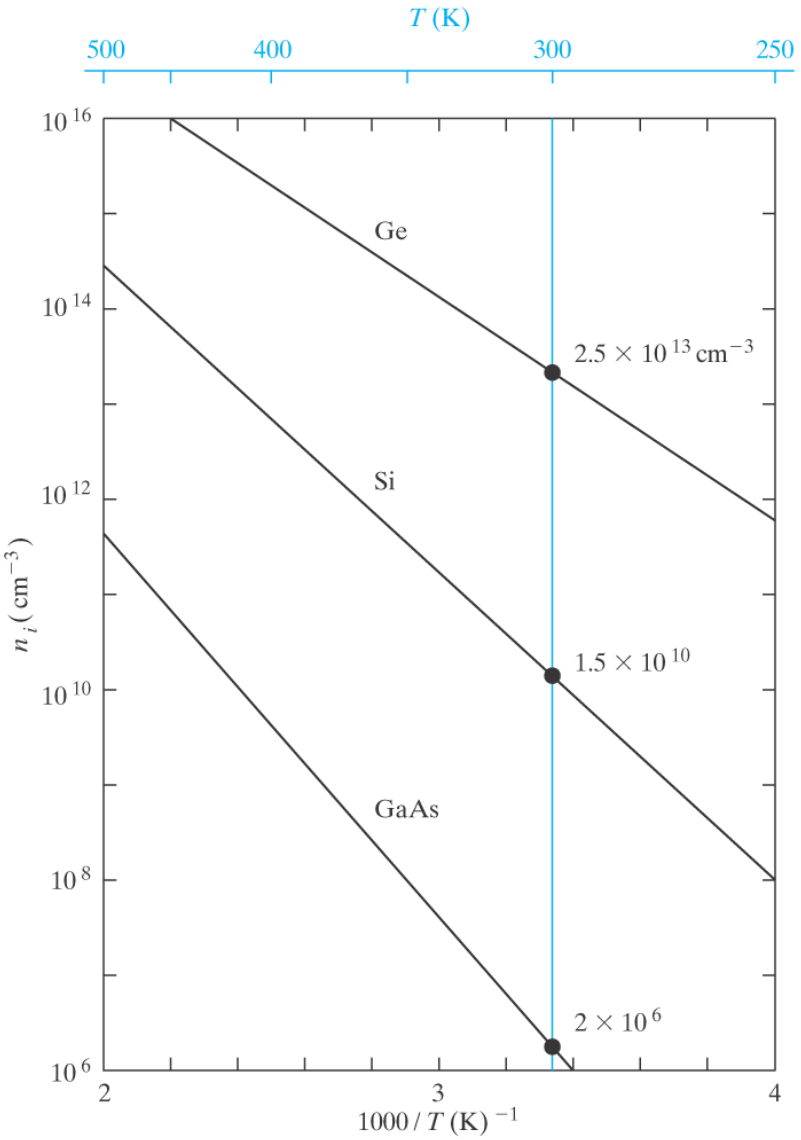
$$p_o = 2.25\times10^{3}\,cm^{-3}$$

$$n_o = n_i(T) \bullet e^{(E_f-E_i)/kT}$$

$$e^{(E_f-E_i)/kT} = \frac{n_o}{n_i}$$

$$E_f - E_i = kT\ln\frac{n_o}{n_i} = 0.407eV$$

Where Did We Get n_i?



Calculate using:

$$n_i = \sqrt{N_c N_v}\, e^{-E_g/2kT}$$

or consult a reference (table, etc.)

Temperature Dependence of Carrier Concentration

Intrinsic Concentration for Si, Ge, and GaAs

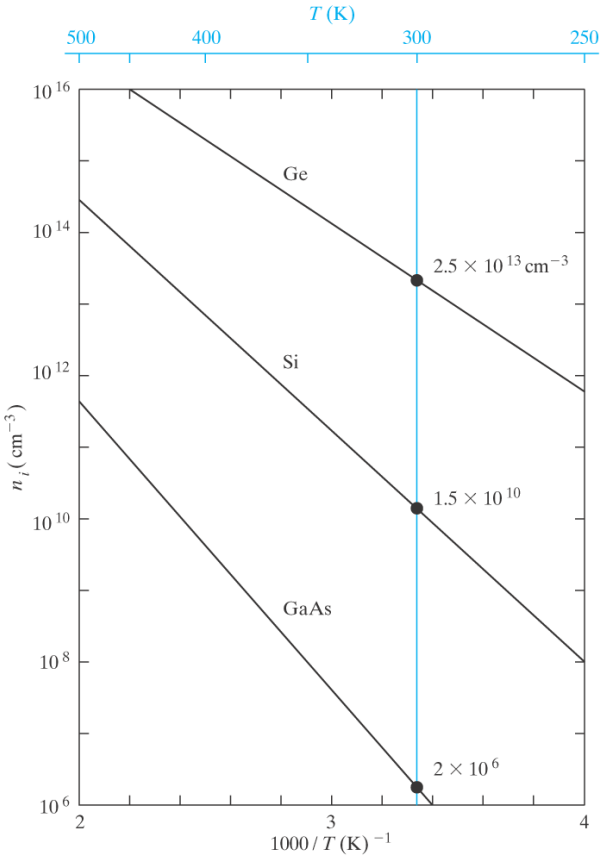


Figure 3.17

Intrinsic carrier concentration for Ge, Si, and GaAs as a function of inverse temperature. The room temperature values are marked for reference.

$$n_o p_o = N_c N_v e^{-(E_g)/kT} = n_i^2(T)$$

$$n_i(T) = \sqrt{N_c N_v}\, e^{-(E_g)/2kT}$$

$$n_i(T) = 2\left(\frac{2\pi kT}{h^2}\right)^{3/2} \left(m_n^* m_p^*\right)^{3/4} e^{-(E_g)/2kT}$$

Intrinsic Carrier Concentration Depends Upon:

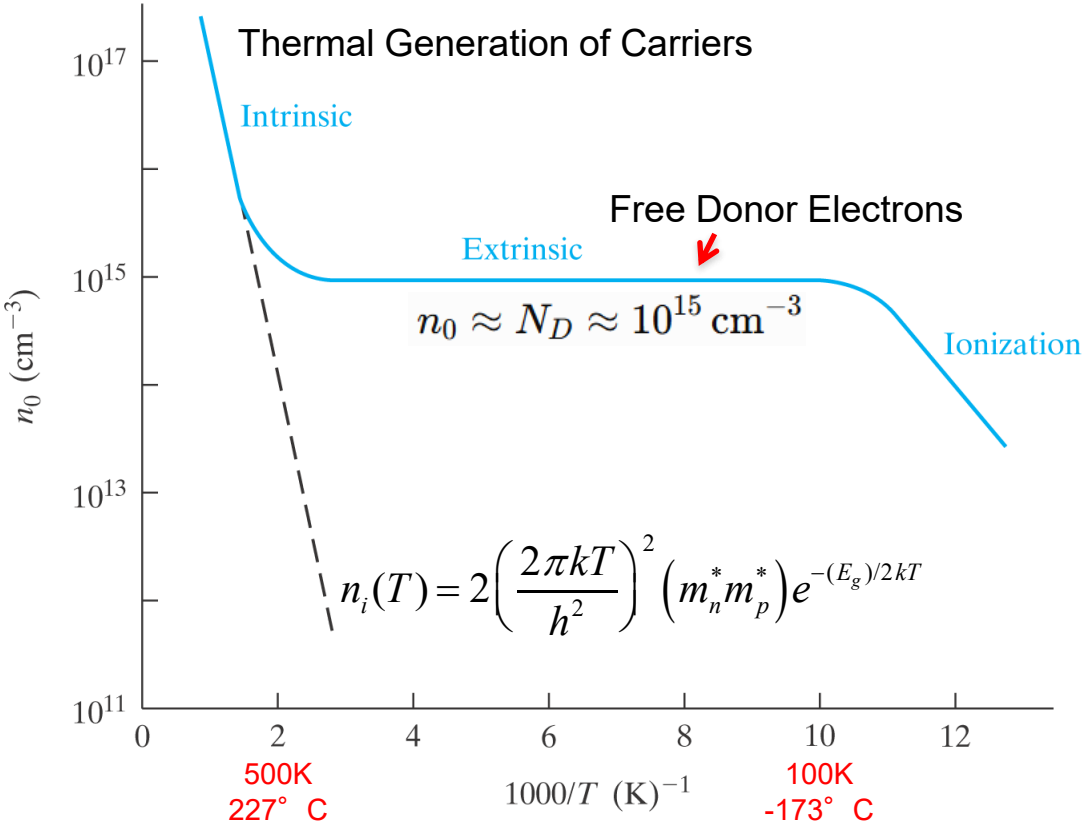
- Energy Gap E_g
- Temperature T
- Electron Effective Mass
- Hole Effective Mass

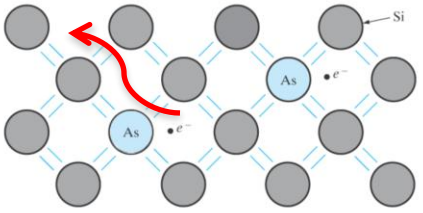
Note: Graph is neglecting $T^{3/2}$ term and $E_g(T)$

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Carrier Concentration Temperature Dependence

Carrier concentration in Si doped with 10^{15} donors/cm^3





$n \propto e^{-E_d/2kT}$

Bound Donor Electrons

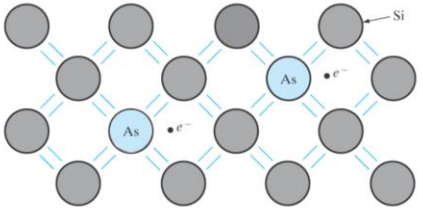


Figure 3.18

Carrier concentration vs. inverse temperature for Si doped with 10^{15} donors/cm^3.

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Compensation and Space Charge Neutrality

What Happens if the Crystal is Doped With Both Donors and Acceptors?

Charge Neutrality Equation :

$$p_o + N_d^+ = n_o + N_a^-$$

N_d^+ : ionized donor (positively charged)
N_a^- : ionized acceptor (negatively charged)

Example Revisited:

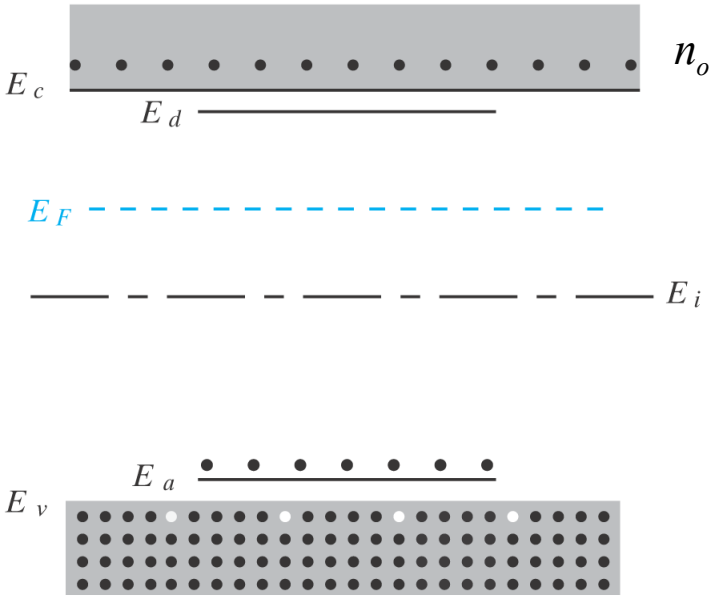
Donor-Doped Silicon, 10^{17} cm^{-3}

Since $N_d(As) >> n_i$,

$$n_o \approx N_d = 10^{17}\,cm^{-3}$$

$$p_o = \frac{n_i^2}{n_o} = \frac{\left(1.5 \times 10^{10}\,cm^{-3}\right)^2}{10^{17}\,cm^{-3}}$$

$$p_o = 2.25 \times 10^{3}\,cm^{-3} << 10^{17}\,cm^{-3}$$



$$n_o = \left(N_d^+ - N_a^-\right) + p_o \approx N_d^+ - N_a^-$$

Figure 3.19

Compensation in an n-type semiconductor ($N_d > N_a$).

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What Happens if the Crystal is Doped With Both Donors and Acceptors?

Charge Neutrality Equation

$$p_o + N_d^+ = n_o + N_a^-$$

Case 1: $N_D > N_A$ and $N_D - N_A >> n_i$

Given GaAs at 300K with $n_{i,GaAs} = 2\times10^{16}\ cm^{-3}$

$N_D = 1\times10^{17}\ cm^{-3}$

$N_A = 5\times10^{16}\ cm^{-3}$

Then:

$$n_o \approx n_o - p_o = N_D^+ - N_A^- = 1\times10^{17} - 5\times10^{16}\ cm^{-3}$$

$$n_o \approx 5\times10^{16}\ cm^{-3}$$

$$p_o = \frac{n_i^2}{n_o} \approx \frac{\left(2\times10^6\right)^2}{5\times10^{16}} = 8\times10^{-5}\ cm^{-3}$$

Case 2: $N_A > N_D$ and $N_A - N_D >> n_i$

Given Si at 300K with $n_{i,Si} = 1.5\times10^{10}\ cm^{-3}$

$N_A = 1\times10^{17}\ cm^{-3}$

$N_D = 5\times10^{16}\ cm^{-3}$

Then:

$$p_o \approx p_o - n_o = N_A^- - N_D^+ = 1\times10^{17} - 5\times10^{16}\ cm^{-3}$$

$$p_o \approx 5\times10^{16}\ cm^{-3}$$

$$n_o = \frac{n_i^2}{p_o} \approx \frac{\left(1.5\times10^{10}\right)^2}{5\times10^{16}} = 4.5\times10^{3}\ cm^{-3}$$

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What Happens Close to the Intrinsic Concentration?

Case 3: $N_D > N_A$ and $N_D - N_A \approx n_i$

Given Ge at 300K with $N_D = 1 \square 10^{14}\, cm^{-3}$ and $N_A = 7 \square 10^{13}\, cm^{-3}$

Then: $n_{i,Ge} = 3 \times 10^{13}\ cm^{-3}$

$$n_o - p_o = N_D^+ - N_A^- = 1 \square 10^{14} - 7 \square 10^{13}\, cm^{-3}$$

$$n_o - p_o = 3 \square 10^{13}\, cm^{-3} \qquad (1)$$

$$p_o = \frac{n_i^2}{n_o} = \frac{\left(2.5 \square 10^{13}\right)^2}{n_o} \qquad (2)$$

Substituting (2) into (1):

$$n_o - \frac{\left(2.5 \square 10^{13}\right)^2}{n_o} = 3 \square 10^{13}\, cm^{-3}$$

Multiply through by n_o and rearrange:

$$n_o^2 - 3 \square 10^{13} n_o - \left(2.5 \square 10^{13}\right)^2 = 0$$

Solution to quadratic equation:

$$n_o = \frac{3 \square 10^{13} \pm \sqrt{\left(3 \square 10^{13}\right)^2 + 4\left(2.5 \square 10^{13}\right)^2}}{2} = \frac{3 \square 10^{13} \pm 5.8 \square 10^{13}}{2} = 4.4 \square 10^{13} \quad \square \quad \text{disgard negative}$$

$$p_o = 1.4 \square 10^{13} \quad \square \quad \text{use either expression to calculate}$$

Charge Neutrality Equation

$$p_o + N_d^+ = n_o + N_a^-$$

Different from n_0–p_0

→ When close to the intrinsic concentration, we cannot neglect these carriers!

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Drift of Carriers in Electric Fields

Conductivity and Mobility

Drift Versus Random Motion

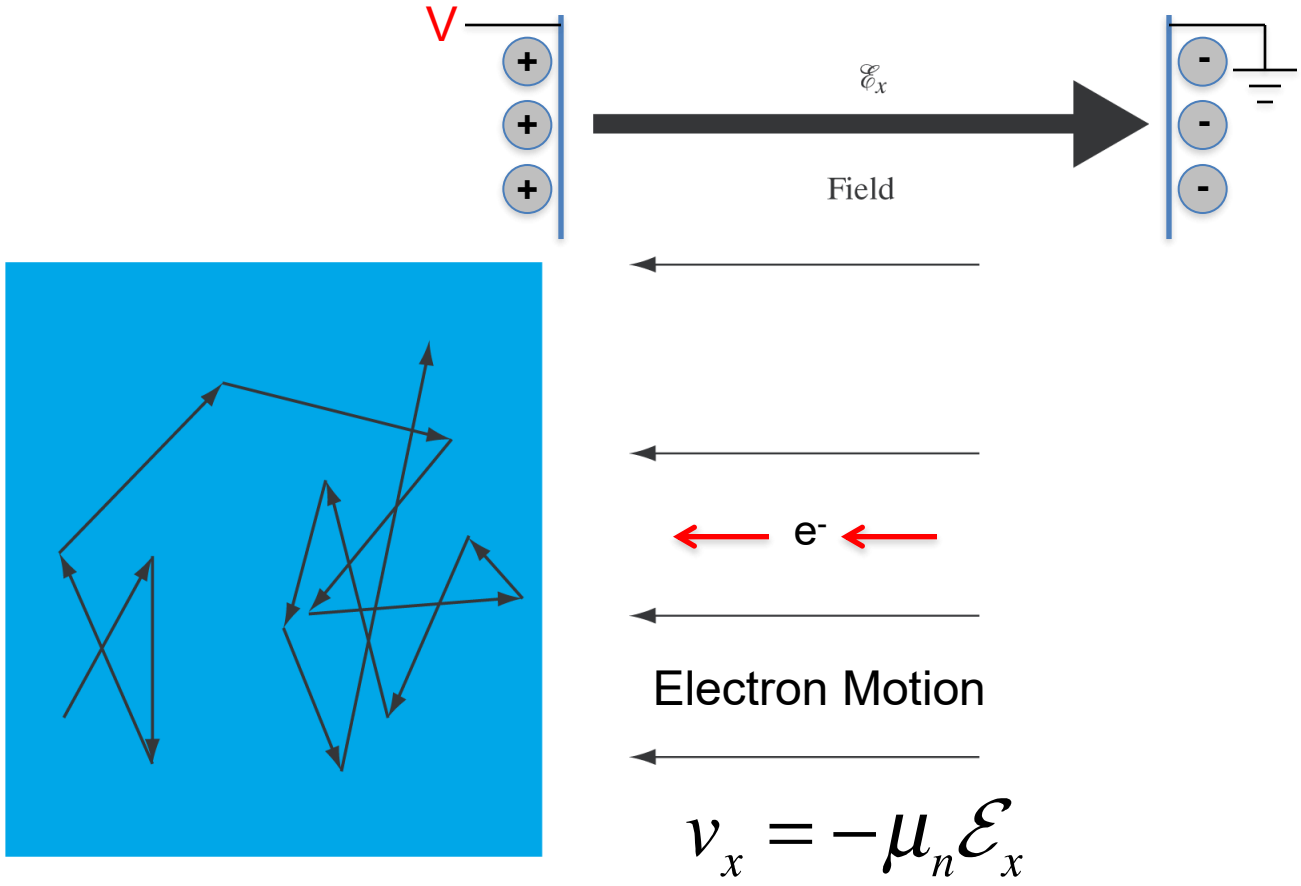


Figure 3.20

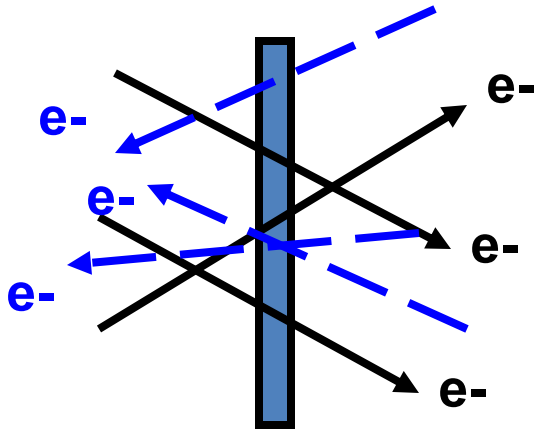
(a) Random thermal motion of an electron in a solid;
(b) well directed drift velocity with an applied electric field.

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Diffusion: A Brief Comment

- In the case of a uniform distribution of carriers in the solid, the random motion is averaged out and there is no net movement of carriers from one region to another
- If there is no net motion of carriers, there is no net current
- We will consider carrier drift under the condition where the distribution of carriers is uniform such that there is no diffusion contribution to the net current
  - We will come back to this assumption later

For Uniform e^- Distribution



$$\text{Net "e}^-\text{" flow} = 0$$

$$\sum_i \bar{e}_i \bullet v_i = 0$$

Drift in a Semiconductor

Assume uniform distribution of carriers: no diffusion

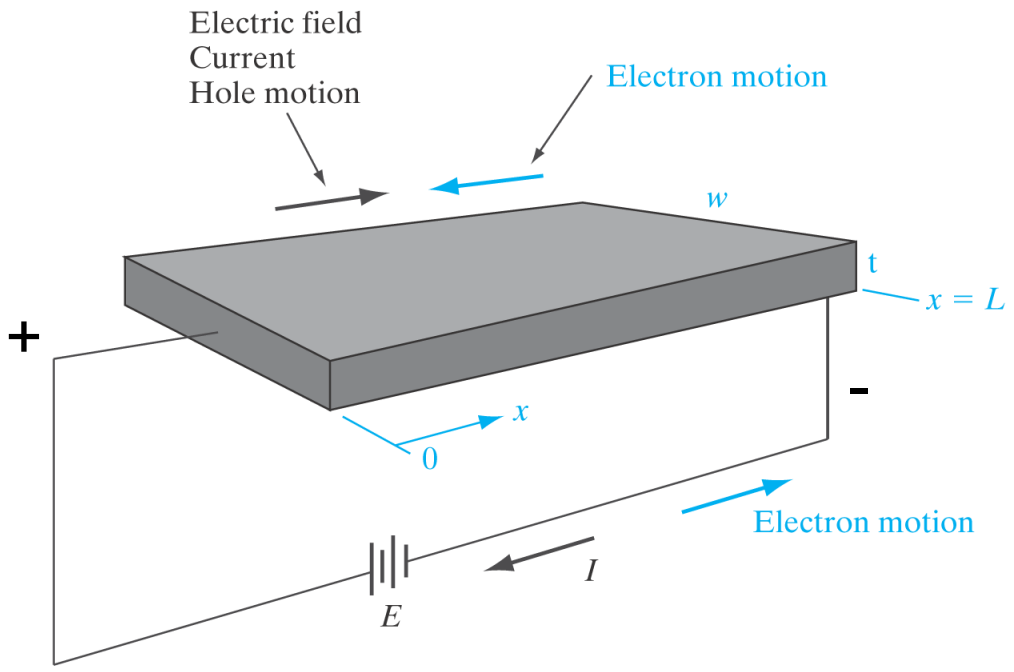


Figure 3.21

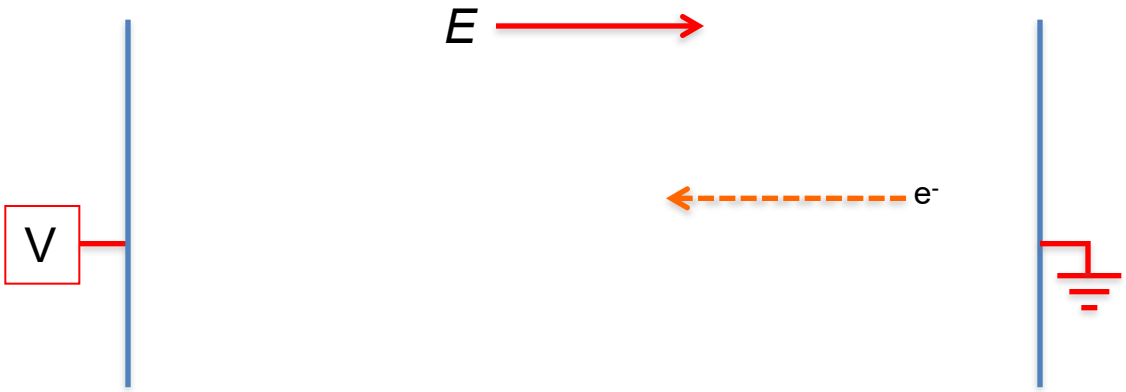
Drift of electrons and holes in a semiconductor bar.

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Electron Motion in a Crystal

Case 1: in vacuum

- An electron in a vacuum will continue to accelerate under the influence of an electric field



Case 2: In crystal lattice

- An electron traveling through the crystal will experience the periodic potential of the lattice atoms
- The electron will be scattered as it encounters + and – regions of space charge
- This will impact a momentum change to the electron
- A faster electron will scatter more quickly, imparting more momentum change per unit time

